Perturb-Seq data handoff

20 single-condition datasets for Open Targets

Perturbation Catalogue

Dataset export contract

2026-10-13

Coverage and record counts

20 datasets **1.661B DEA rows** **7.286M GSEA rows**

Study set	Dataset IDs	N	DEA rows	GSEA rows
Adamson	`adamson_2016_pilot`; `adamson_2016_upr_perturb_seq`	2	1.108M	4.27K
Norman	`norman_2019_raw`	1	1.345M	5.10K
Nadig	`nadig_2025_hepg2`; `nadig_2025_jurkat`	2	60.402M	229.97K

Replogle	`replogle_2022_k562_essential_normalized`
	`replogle_2022_rpe1_essential_normalized`
	`replogle_2022_k562_gw_normalized`
		3	174.930M	669.66K

Zhu	`zhu_2025_D{1,2,3,4}_{rest,stim8hr,stim48hr}_cl`	12	1.423B	6.377M
Total	All 20 IDs above	20	1.661B	7.286M

Counts are from the dev data snapshot queried on 2026-10-13. Zhu brace notation expands to three IDs for each of D1–D4.

Metadata JSON: study and dataset context

Common fields

- Identity: `dataset_id`, study title, year and authors
- Context: modalities, perturbation, tissue, cell type and cell line
- Scale: target and sample counts
- Full JSON also has assay, ontology, license and processing fields

Repeated annotations and counts are JSON arrays.

Example record excerpt


```
{
  "dataset_id": "adamson_2016_pilot",
  "study_title": "A Multiplexed Single-Cell CRISPR
     Screening Platform Enables Systematic Dissection of
      the Unfolded Protein Response",
  "study_year": "2016",
  "first_author": "Britt Adamson",
  "data_modalities": ["Perturb-seq"],
  "perturbation_type_labels": ["CRISPRi"],
  "cell_type_labels": ["lymphoblast"],
  "cell_line_labels": ["K 562 cell"],
  "number_of_perturbed_targets": ["8"],
  "number_of_perturbed_samples": ["5752"]
}
```


Values are an excerpt from the dataset metadata record; count fields retain their source array representation.

DEA Parquet: one row per target–effect-gene pair

Column	Type	Example value: adamson_2016_pilot
perturbed_target_ensg	STRING	ENSG00000134107
perturbed_target_name	STRING	BHLHE40
effect_gene_ensg	STRING	ENSG00000094914
effect_gene_name	STRING	AAAS
log2foldchange	FLOAT64	0.0722239017
padj	FLOAT64	0.9998721657
score_name	STRING	Wilcoxon Score
score_value	FLOAT64	0.4150409698
cell_type	STRING, nullable	null

Record interpretation

The perturbed target is the intervention; the effect gene is the measured transcript. The per-dataset file omits `dataset_id`; the filename supplies it.

GSEA Parquet: enrichment by target and gene set

Column	Type	Example value: adamson_2016_pilot
perturbed_target_ensg	STRING	ENSG00000066336
perturbed_target_symbol	STRING	SPI1
term	STRING	HALLMARK_ALLOGRAFT_REJECTION
es	FLOAT64	-0.3599006142
nes	FLOAT64	-1.1792592227
pval	FLOAT64	0.1935483871
sidak	FLOAT64	0.9999593387
fdr	FLOAT64	0.3540276454
geneset_size	INT64	81
leading_edge	ARRAY<STRING>	[SPI1, NPM1, RPS9, CTSS, RPS19, ...]
cell_type	STRING, nullable	null

Record interpretation

leading_edge is a typed string array in Parquet. In the CSV.GZ companion it is represented as a semicolon-separated list.

Per-dataset artifact names

Paths are relative to the release bucket; an optional version prefix may precede `perturb-seq/`.

Artifact	Object name pattern
Metadata	`perturb-seq/<dataset_id>.metadata.json`
DEA data	`perturb-seq/<dataset_id>.parquet`
DEA CSV mirror	`perturb-seq/<dataset_id>.csv.gz`
GSEA data	`perturb-seq/<dataset_id>.gsea.parquet`
GSEA CSV mirror	`perturb-seq/<dataset_id>.gsea.csv.gz`

- One metadata JSON plus DEA and GSEA files for each of the 20 dataset IDs.
- Parquet preserves typed columns; CSV.GZ uses the matching human-readable field labels.
- The metadata object contains the join key; DEA and GSEA rows are scoped by their per-dataset filenames.